



What's proven, what we checked, what isn't



PROVEN • PUBLISHED

- Physics: rain weakens microwave signals (ITU-R P.838)
- Method: rain from a phone network (Science 2006)
- Scale: country-wide rain maps (PNAS 2013)
- Open-source tools exist (pycomlink, KIT Germany)



OUR CHECK • PUBLIC DATA

- 500 real links, Germany, 11 days, radar as reference
- 0.95 network-wide match; 0.88 typical link; 0.78 map level
- Rain / no-rain days: 91% detected, 5% false
- Links alone failed all three accuracy gates: 68% of trigger events ($\geq 80\%$), totals 18% low ($\pm 10\%$), 18% unconfirmed ($< 15\%$)



NOT YET PROVEN • THE PILOT

- Jio's usable coverage → Gate 1
- Accuracy in Indian monsoon → Gate 2
- Business value inside Reliance → Gate 3
- Who pays → payer test in the pilot
- Insurance-grade triggers → accuracy + actuarial gates, shadow payouts

We know what is proven. We know what isn't. The pilot is built around the unknowns.



Every limit we know of, and our answer

Limitation	Our answer
Jio may not have enough usable microwave links	Coverage audit is Gate 1; radar, gauges and satellite still work without links
Fibre doesn't sense rain	We only use microwave links
A link measures path-averaged rain, not a point	Fusion with radar and gauges turns it into grid estimates
Links alone failed all three accuracy gates	We never use links alone; accuracy picks the mix
18% of our triggers weren't confirmed by radar	Multi-source confirmation and local calibration
Rain can be confused with equipment faults	Network-health flags (faults, outages, maintenance) and cross-link checks
Performance in India is unknown	Back-test Jio's archived Mumbai monsoon data (Gate 2)
1 km resolution isn't guaranteed	Resolution depends on link density and fusion; reported per area
Platforms' willingness to pay is unknown	Prove value inside Reliance first (Gate 3, matched zones), then sell
Insurance basis risk	Small zones, multi-source trigger, review process
₹300 payout is arbitrary today	Prototype assumption; set by actuarial calibration
₹208 isn't priced on the 10 mm in 3 h trigger	Priced on 25 years of daily IMD risk; the pilot recalibrates the trigger
Towers don't sense heat	IMD stations and satellite feed the same engine
Insurance may never launch	The intelligence business stands on its own



The questions we expect, and short answers

If a judge asks	Say
"How closely did links track radar?" / "Why only 68%?"	Network-wide hourly match 0.95; a typical single link 0.88; map level 0.78. The trigger test is harsher: a 9 mm estimate against an 11 mm event is a miss, and links read 18% low. So 68%.
"Mumbai already has hyperlocal weather."	Exactly: BMC gauges and IMD radar are inputs we fuse, not rivals. The pilot's "radar + gauges, no links" run measures what Jio's links add.
"Jio already knows rain fade."	It may. Then Gate 1 confirms cheaply what weather information is available. If it doesn't, the same telemetry is a new signal.
"What if it doesn't rain in days 61-90?"	The back-test replays archived monsoon events, including network-ops value. The full live test runs in Mumbai's 2027 monsoon (months 4-12).
"Why ₹208 if the live trigger is 10 mm in 3 h?"	₹208 isn't priced from that prototype trigger. It uses 25 years of daily IMD heavy-rain history, capped at five paid days. The pilot measures the 3-hour trigger's real frequency and recalibrates before launch.
"Did you build the model?"	We didn't invent the method; it's published research. We implemented it with open-source tools, ran it ourselves on real link data, and measured where it fails.
"Germany isn't India."	Agreed. Gate 2 back-tests Jio's archived Mumbai monsoon data against held-out IMD and BMC gauges. Nothing scales until it passes.
"Who decides whether Jio's own system pays?"	Jio only supplies the trigger data. A licensed insurer owns the contract, the rules and the payout; the trigger is multi-source and auditable.